

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF TECHNOLOGY AND ENGINEERING

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Subject Name: Computer Networks

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Lab Manual

1	Study of basic network command and Network configuration commands. 1) ping : - Helps in determining TCP/IP Networks IP address as well as to verify if the device with that IP address is connected to the pc or not - Determining issues with the network and assist in resolving them. <pre>C:\Users\Admin>ping 172.16.0.1 Pinging 172.16.0.1 with 32 bytes of data: Reply from 172.16.0.1: bytes=32 time<1ms TTL=64 Reply from 172.16.0.1: bytes=32 time<1ms TTL=64 Reply from 172.16.0.1: bytes=32 time<1ms TTL=64 Reply from 172.16.0.1: bytes=32 time<1ms TTL=64 Ping statistics for 172.16.0.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\Users\Admin>ping www.google.com Pinging www.google.com [216.58.199.132] with 32 bytes of data: Reply from 216.58.199.132: bytes=32 time=15ms TTL=56 Reply from 216.58.199.132: bytes=32 time=14ms TTL=56 Reply from 216.58.199.132: bytes=32 time=11ms TTL=56 Reply from 216.58.199.132: bytes=32 time=16ms TTL=56 Ping statistics for 216.58.199.132: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 11ms, Maximum = 16ms, Average = 14ms</pre> 2) Tracert : - Used to visually see a packet being sent and received and amount of hops required for that packet to get to its destination
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Prepared By:

```

C:\Users\Admin>tracert www.google.com

Tracing route to www.google.com [216.58.199.132]
over a maximum of 30 hops:

  1  <1 ms    <1 ms    <1 ms    172.16.0.1
  2   6 ms     5 ms     5 ms     172.24.195.242
  3   *        16 ms    *        218.248.235.198
  4  12 ms     13 ms    11 ms    74.125.48.138
  5   *        *        *        Request timed out.
  6  12 ms     10 ms    13 ms    bom07s01-in-f4.1e100.net [216.58.199.132]

Trace complete.

```

3) ipconfig :

- displays the network settings currently assigned to any or all the network adapters in the machine
- specially used to verify network connection as well as to verify network settings

```

C:\Users\Admin>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::9d78:5a25:73b6:ee99%14
    IPv4 Address. . . . . : 172.16.3.115
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 172.16.0.1

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::5e1:150b:2628:6c98%21
    IPv4 Address. . . . . : 192.168.154.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::e93a:11fb:2a07:914a%22
    IPv4 Address. . . . . : 192.168.75.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter VirtualBox Host-Only Network:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::d8b8:dc46:46c:ec5%23
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Tunnel adapter isatap.{A5CAA4C8-D1F7-4833-AB1E-A5FDF177C4AD}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter isatap.{EF2E66F6-A99C-4F62-A316-F9EED199D470}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter isatap.{ED558BA6-353D-4949-AF0A-4ACA592E9BED}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter isatap.{548D23E2-6139-4A67-AE2C-207D1C2AD6AD}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter isatap.{F4A6B41A-A972-43E3-8EB3-11590F6B93D0}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

```

4) Pathping :

- Provides information about network latency and network loss at intermediate hops between source and destination

- Calculates packet loss in every intermediate routers
- It sends echo requests via ICMP(Internet Control Message Protocol) and analyzing the result

```
C:\Users\Admin>pathping www.google.com

Tracing route to www.google.com [216.58.199.132]
over a maximum of 30 hops:
  0  4046-A-15 [172.16.3.115]
  1  172.16.0.1
  2  172.24.195.242
  3  * * *
Computing statistics for 50 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
  0             Lost/Sent = Pct  Lost/Sent = Pct  4046-A-15 [172.16.3.115]
  1      0ms      0/ 100 = 0%      0/ 100 = 0%      172.16.0.1
  2     15ms      3/ 100 = 3%      0/ 100 = 0%      172.24.195.242
Trace complete.

C:\Users\Admin>pathping -n www.google.com

Tracing route to www.google.com [216.58.199.132]
over a maximum of 30 hops:
  0  172.16.3.115
  1  172.16.0.1
  2  172.24.195.242
  3  218.248.235.198
  4  74.125.48.138
  5  * * *
Computing statistics for 100 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
  0             Lost/Sent = Pct  Lost/Sent = Pct  172.16.3.115
  1      0ms      0/ 100 = 0%      0/ 100 = 0%      172.16.0.1
  2     17ms      1/ 100 = 1%      0/ 100 = 0%      172.24.195.242
  3     11ms      1/ 100 = 1%      0/ 100 = 0%      218.248.235.198
  4     13ms      1/ 100 = 1%      0/ 100 = 0%      74.125.48.138
Trace complete.
```

5) Nslookup :

- Displays information that can be used to diagnose DNS infrastructure

```
C:\Users\Admin>nslookup 117.239.83.200
Server:  Unknown
Address:  172.16.0.1

Name:     static.ill.117.239.83.200/24.bsnl.in
Address:  117.239.83.200
```

6) Netstat :

- Displays active TCP connections
- Ports on which the computer is listening
- Ethernet statistics
- IP routing Table
- IPv4 statistics
- IPv6 statistics

```

C:\Users\Admin>netstat
Active Connections

Proto Local Address           Foreign Address         State
TCP   127.0.0.1:49165          4046-A-15:49166        ESTABLISHED
TCP   127.0.0.1:49166          4046-A-15:49165        ESTABLISHED
TCP   127.0.0.1:49170          4046-A-15:49171        ESTABLISHED
TCP   127.0.0.1:49171          4046-A-15:49170        ESTABLISHED
TCP   127.0.0.1:49174          4046-A-15:49175        ESTABLISHED
TCP   127.0.0.1:49175          4046-A-15:49174        ESTABLISHED
TCP   127.0.0.1:49480          4046-A-15:49481        ESTABLISHED
TCP   127.0.0.1:49481          4046-A-15:49480        ESTABLISHED
TCP   127.0.0.1:56050          4046-A-15:56051        ESTABLISHED
TCP   127.0.0.1:56051          4046-A-15:56050        ESTABLISHED
TCP   127.0.0.1:56970          4046-A-15:56971        ESTABLISHED
TCP   127.0.0.1:56971          4046-A-15:56970        ESTABLISHED
TCP   127.0.0.1:56975          4046-A-15:56974        TIME_WAIT
TCP   172.16.3.115:8194        sophos-endpoint:52411   ESTABLISHED
TCP   172.16.3.115:49168        sophos-endpoint:8194    ESTABLISHED
TCP   172.16.3.115:49338        1.2.3.4:9922           ESTABLISHED
TCP   172.16.3.115:49482        ec2-34-199-252-227:https ESTABLISHED
TCP   172.16.3.115:50832        sophos-endpoint:8194    ESTABLISHED
TCP   172.16.3.115:56052        bom05s10-in-f10:https   ESTABLISHED
TCP   172.16.3.115:56973        bom07s15-in-f3:https    ESTABLISHED
TCP   172.16.3.115:57159        admin-PC:ms-sql-s        SYN_SENT
TCP   192.168.75.1:8194         4046-A-15:49172        ESTABLISHED
TCP   192.168.75.1:49172        4046-A-15:8194          ESTABLISHED
TCP   [fe80::9d78:5a25:73b6:ee99%14]:57054 admin-PC:microsoft-ds

```

7) Arp (Address Routing Protocol) :

- Displays, adds, and removes ARP information from network devices

Arp-a

- Displays current Arp entries by interrogating the current protocol data

```

C:\Users\Admin>arp -a

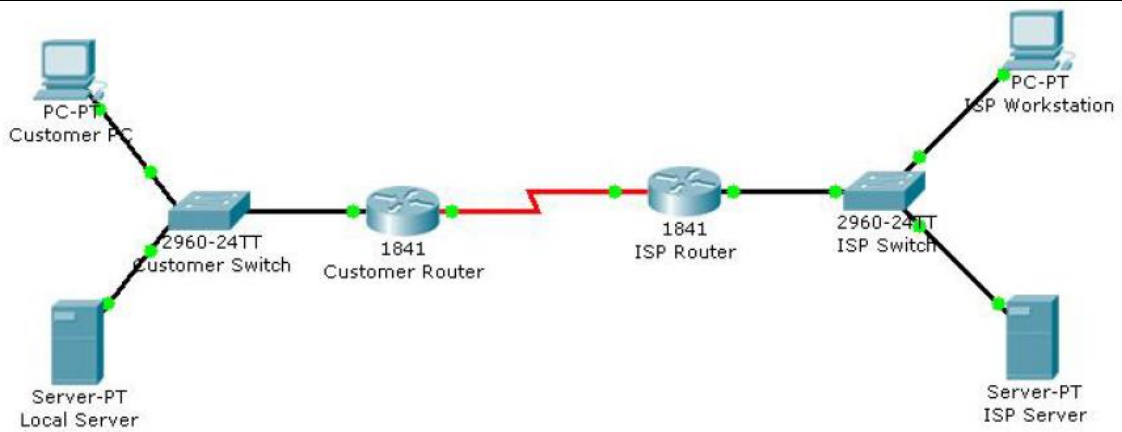
Interface: 172.16.3.115 --- 0xe
Internet Address      Physical Address        Type
172.16.0.1            00-1a-8c-6b-76-ac       dynamic
172.16.1.177          74-46-a0-92-fe-1d       dynamic
172.16.2.54           6c-3b-e5-33-c4-8a       dynamic
172.16.2.78           40-a8-f0-41-d1-e1       dynamic
172.16.2.83           a0-d3-c1-2d-76-22       dynamic
172.16.2.89           40-a8-f0-41-d1-97       dynamic
172.16.2.249          6c-3b-e5-3e-c3-19       dynamic
172.16.3.108          e0-69-95-b2-c4-a4       dynamic
172.16.3.132          e0-69-95-b2-c5-34       dynamic
172.16.3.138          e0-69-95-b2-c2-d3       dynamic
172.16.3.146          e0-69-95-b2-c2-cb       dynamic
172.16.3.148          e0-69-95-b2-c4-a0       dynamic
172.16.3.248          6c-3b-e5-38-13-3b       dynamic
172.16.6.166          6c-3b-e5-33-f8-9f       dynamic
172.16.6.248          6c-3b-e5-38-12-ed       dynamic
172.16.11.75          00-50-56-95-7f-f4       dynamic
172.16.12.26          a0-d3-c1-15-b0-38       dynamic
172.16.12.71          10-78-d2-52-0a-e5       dynamic
172.16.12.215         6c-3b-e5-38-13-1f       dynamic
172.16.12.228         6c-3b-e5-2a-74-1a       dynamic
172.16.13.74          a0-d3-c1-2d-77-6d       dynamic
172.16.13.103         6c-3b-e5-32-b3-8a       dynamic
172.16.15.4           00-01-6c-44-2b-60       dynamic

Interface: 192.168.154.1 --- 0x15
Internet Address      Physical Address        Type
192.168.154.255       ff-ff-ff-ff-ff-ff       static
224.0.0.2             01-00-5e-00-00-02       static
224.0.0.22            01-00-5e-00-00-16       static
224.0.0.251           01-00-5e-00-00-fb       static
224.0.0.252           01-00-5e-00-00-fc       static
239.255.255.250       01-00-5e-7f-ff-fa       static

Interface: 192.168.75.1 --- 0x16
Internet Address      Physical Address        Type
192.168.75.255        ff-ff-ff-ff-ff-ff       static
224.0.0.2             01-00-5e-00-00-02       static
224.0.0.22            01-00-5e-00-00-16       static
224.0.0.251           01-00-5e-00-00-fb       static
224.0.0.252           01-00-5e-00-00-fc       static
239.255.255.250       01-00-5e-7f-ff-fa       static

Interface: 192.168.56.1 --- 0x17
Internet Address      Physical Address        Type
192.168.56.255        ff-ff-ff-ff-ff-ff       static
224.0.0.2             01-00-5e-00-00-02       static
224.0.0.22            01-00-5e-00-00-16       static
224.0.0.251           01-00-5e-00-00-fb       static
224.0.0.252           01-00-5e-00-00-fc       static
239.255.255.250       01-00-5e-7f-ff-fa       static

```



Objectives

- ☐ Perform an initial configuration of a Cisco Catalyst 2960 switch.

Background / Preparation

In this activity, you will configure these settings on the customer Cisco Catalyst 2960 switch: ☐

Host name ☐ Console password

☐ vty password

☐ Privileged EXEC mode password

☐ Privileged EXEC mode secret ☐ IP address on VLAN1 interface ☐ Default gateway

Note: Not all commands are graded by Packet Tracer.

Step 1: Configure the switch host name.

a. From the Customer PC, use a console cable and terminal emulation software to connect to the console

of the customer Cisco Catalyst 2960 switch.

b. Set the host name on the switch to CustomerSwitch using these commands.

Switch>enable

Switch#configure terminal

Switch(config)#hostname CustomerSwitch

Step 2: Configure the privileged mode password and secret.

a. From global configuration mode, configure the password as cisco.

```
CustomerSwitch(config)#enable password cisco
```

b. From global configuration mode, configure the secret as cisco123.

```
CustomerSwitch(config)#enable secret cisco123
```

Step 3: Configure the console password.

a. From global configuration mode, switch to configuration mode to configure the console line.

```
CustomerSwitch(config)#line console 0
```

b. From line configuration mode, set the password to cisco and require the password to be entered at login.

```
CustomerSwitch(config-line)#password cisco
```

```
CustomerSwitch(config-line)#login
```

```
CustomerSwitch(config-line)#exit
```

Step 4: Configure the vty password.

a. From global configuration mode, switch to the configuration mode for the vty lines 0 through 15.

```
CustomerSwitch(config)#line vty 0 15
```

b. From line configuration mode, set the password to cisco and require the password to be entered at login.

```
CustomerSwitch(config-line)#password cisco
```

```
CustomerSwitch(config-line)#login
```

```
CustomerSwitch(config-line)#exit
```

Step 5: Configure an IP address on interface VLAN1.

From global configuration mode, switch to interface configuration mode for VLAN1, and assign the IP address

192.168.1.5 with the subnet mask of 255.255.255.0.

CustomerSwitch(config)#interface vlan 1

CustomerSwitch(config-if)#ip address 192.168.1.5 255.255.255.0

CustomerSwitch(config-if)#no shutdown

CustomerSwitch(config-if)#exit

Step 6: Configure the default gateway.

a. From global configuration mode, assign the default gateway to 192.168.1.1.

CustomerSwitch(config)#ip default-gateway 192.168.1.1

b. Click the Check Results button at the bottom of this instruction window to check your work.

Step 7: Verify the configuration.

The Customer Switch should now be able to ping the ISP Server at 209.165.201.10. The first one or two pings

may fail while ARP converges.

CustomerSwitch(config)#end

CustomerSwitch#ping 209.165.201.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 209.165.201.10, timeout is 2 seconds:

..!!!

Success rate is 60 percent (3/5), round-trip min/avg/max = 181/189/197 ms

CustomerSwitch#

Reflection

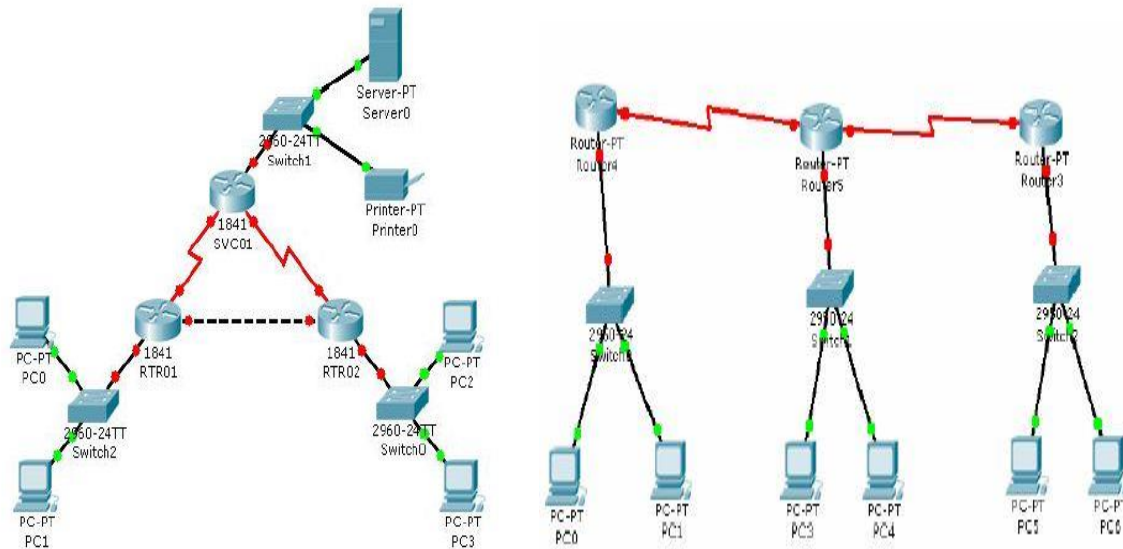
a. What is the significance of assigning the IP address to the VLAN1 interface instead of any of the Fast

Ethernet interfaces?

b. What command is necessary to enforce password authentication on the console and vty lines?

c. How many gigabit ports are available on the Cisco Catalyst 2960 switch that you used in the activity?

3 Routing Static vs Dynamic(RIP OSPF)



Objectives

☐ Configure routers using basic interface configuration commands. ☐ Enable RIP. ☐ Verify the RIP configuration.

Background / Preparation

A simple routed network has been set up to assist in reviewing RIP routing behavior. In this activity, you will

configure RIP across the network and set up end devices to communicate on the network.

Step 1: Configure the SVC01 router and enable RIP.

a. From the CLI, configure interface Fast Ethernet 0/0 using the IP address 10.0.0.254 /8.

b. Configure interface serial 0/0/0 using the first usable IP address in network 192.168.1.0 /24 to connect to the RTR01 router. Set the clock rate at 64000.

c. Configure interface serial 0/0/1 using the first usable IP address in network 192.168.2.0 /24 with a clock rate of 64000.

d. Using the no shutdown command, enable the configured interfaces.

e. Configure RIP to advertise the networks for the configured interfaces.

f. Configure the end devices.

i. Server0 uses the first usable IP address in network 10.0.0.0 /8. Specify the appropriate default gateway and subnet mask.

ii. Printer0 uses the second usable IP address in network 10.0.0.0 /8. Specify the appropriate default gateway and subnet mask.

Step 2: Configure the RTR01 router and enable RIP.

a. Configure interface Fast Ethernet 0/0 using the first usable IP address in network 192.168.0.0 /24 to connect to the RTR02 router.

b. Configure interface serial 0/0/0 using the second usable IP address in network 192.168.1.0 /24 to connect to the SVC01 router.

c. Configure interface Fast Ethernet 0/1 using the IP address 172.16.254.254 /16.

d. Using the no shutdown command, enable the configured interfaces.

e. Configure RIP to advertise the networks for the configured interfaces.

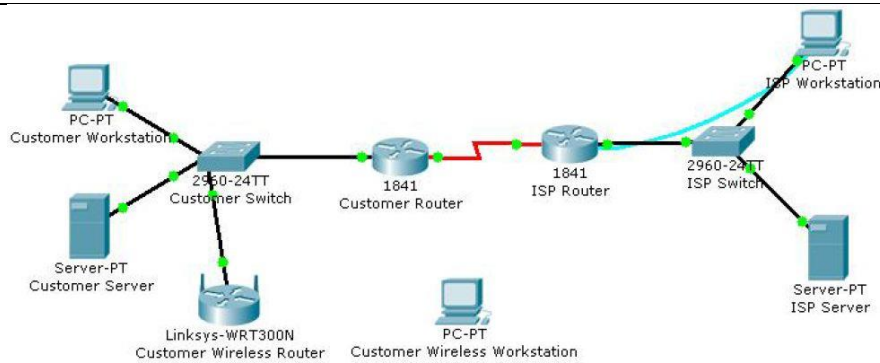
f. Configure the end devices.

i. PC0 uses the first usable IP addresses in network 172.16.0.0 /16.

ii. PC1 uses the second usable IP address in network 172.16.0.0 /16.

iii. Specify the appropriate default gateway and subnet mask on each PC.

	Step 3: Configure the RTR02 router and enable RIP. - Configure interface Fast Ethernet 0/0 using the second usable IP address in network 192.168.0.0 /24 to connect to the RTR01 router. - Configure interface serial 0/0/0 using the second usable IP address in network 192.168.2.0 /24 to connect to the SVC01 router. - Configure interface Fast Ethernet 0/1 using the IP address 172.17.254.254 /16. - Using the no shutdown command, enable the configured interfaces. - Configure RIP to advertise the networks for the configured interfaces. - Configure the end devices. - PC2 uses the first usable IP addresses in network 172.17.0.0 /16. - PC3 uses the second usable IP address in network 172.17.0.0 /16. - Specify the appropriate default gateway and subnet mask on each PC. Step 4: Verify the RIP configuration on each router. - At the command prompt for each router, issue the commands show ip protocols and show ip route to verify RIP routing is fully converged. The show ip protocols command displays the networks the router is advertising and the addresses of other RIP routing neighbors. The show ip route command output displays all routes known to the local router including the RIP routes which are indicated by an "R". - Every device should now be able to successfully ping any other device in this activity. - Click the Check Results button at the bottom of this instruction window to check your work.
4	Configuring WEP on a Wireless Router



Objectives

- Configure WEP security between a workstation and a Linksys wireless router.

Background / Preparation

You have been asked to go back to a business customer and install a new Linksys wireless router for the customer office. The company has some new personnel who will be using wireless computers to save money on

adding additional wired connections to the building. The business is concerned about the security of the

network because they have financial and highly classified data being transmitted over the network.

Your job is

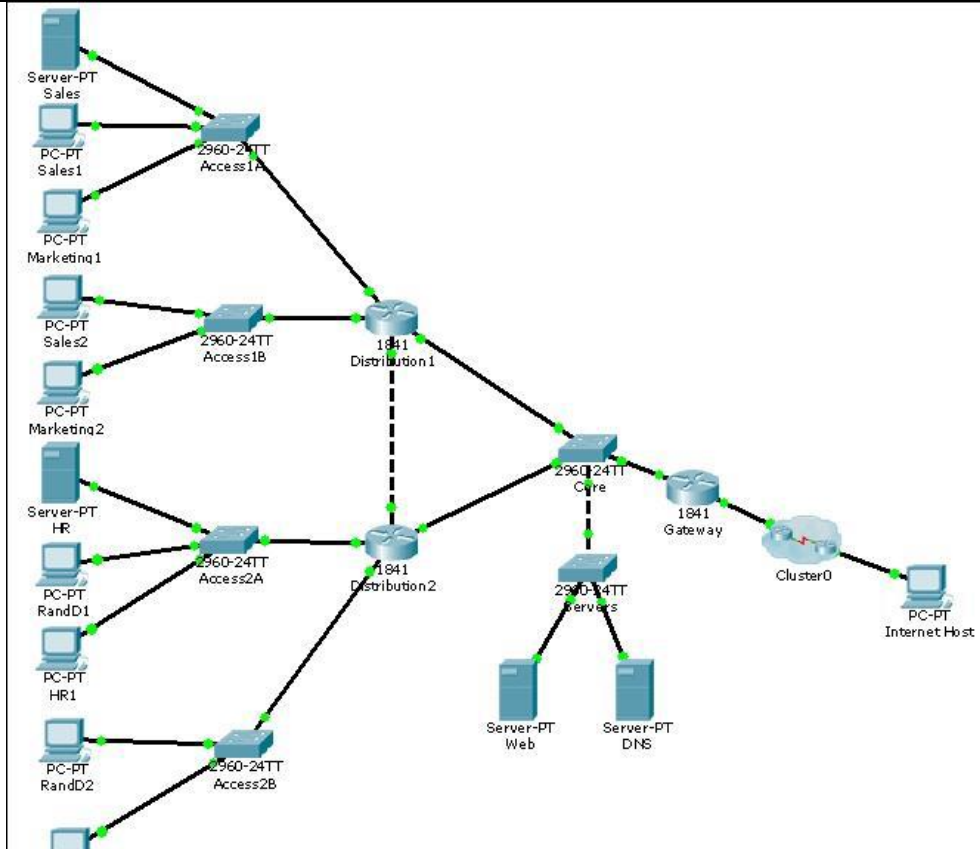
to configure the security on the router to protect the data.

In this activity, you will configure WEP security on both a Linksys wireless router and a workstation.

Step 1: Configure the Linksys wireless router to require WEP.

- a. Click the Customer Wireless Router icon. Then, click the GUI tab to access the router web management interface.
- b. Click the Wireless menu option and change the Network Name (SSID) from Default to CustomerWireless. Leave the other settings with their default options.
- c. Click the Save Settings button at the bottom of the Basic Wireless Settings window.
- d. Click the Wireless Security submenu under the Wireless menu to display the current wireless security parameters.
- e. From the Security Mode drop-down menu, select WEP. f. In the Key1 text box, type 1a2b3c4d5e. This will be the new WEP pre-shared key to access the wireless network.
- g. Click the Save Settings button at the bottom of the Wireless Security window.

	Step 2: Configure WEP on the customer wireless workstation. - Click the Customer Wireless Workstation. - Click the Config tab. - Click the Wireless button to display the current wireless configuration settings on the workstation. - Change the SSID to CustomerWireless. - Change the Security Mode to WEP. Enter 1a2b3c4d5e in the Key text box, and then close the window. Step 3: Verify the configuration. After you configure the correct WEP key and SSID on the customer wireless workstation, notice that there is a wireless connection between the workstation and the wireless router. - Click the Customer Wireless Workstation. - Click the Desktop tab to view the applications that are available. - Click on the Command Prompt application to bring up the command prompt. - Type ipconfig /all and press Enter to view the current network configuration settings. - Type ping 192.168.2.1 to verify connectivity to the LAN interface of the customer wireless router. - Close the command prompt window. - Open a web browser. - In the address bar of the web browser window, type http://192.168.1.10. Press Enter. The Intranet web page that is running on the customer server appears. You have just verified that the customer wireless workstation has connectivity to the rest of the customer network. - Click the Check Results button at the bottom of this instruction window to check your work. Reflection - What is the purpose of using WEP on a wireless network? - What is the significance of the key that you used to secure WEP? - Is WEP the best choice for wireless security?
5	Placing ACLs



Objectives ☐ Verify network connectivity

☐ Examine the Access Control Lists (ACLs) that are configured on the routers ☐ Determine the appropriate interface to apply the ACLs ☐ Examine the effects of the ACL

Background / Preparation

This activity demonstrates how the flow of network traffic is affected by applying an ACL to permit or deny traffic in the network. The network administrator has decided that all external web traffic goes only

to the Web server. Also, in order to protect the data of their employees, the HR server is only accessible to

HR employees. Therefore, ACLs will need to be implemented on the network. Another network technician has already configured the necessary ACLs on both the Gateway and Distribution2 routers.

However, the ACLs have not been applied to an interface. You have been asked to apply the ACLs and verify that the appropriate traffic is permitted or denied.

Required file: Placing ACLs

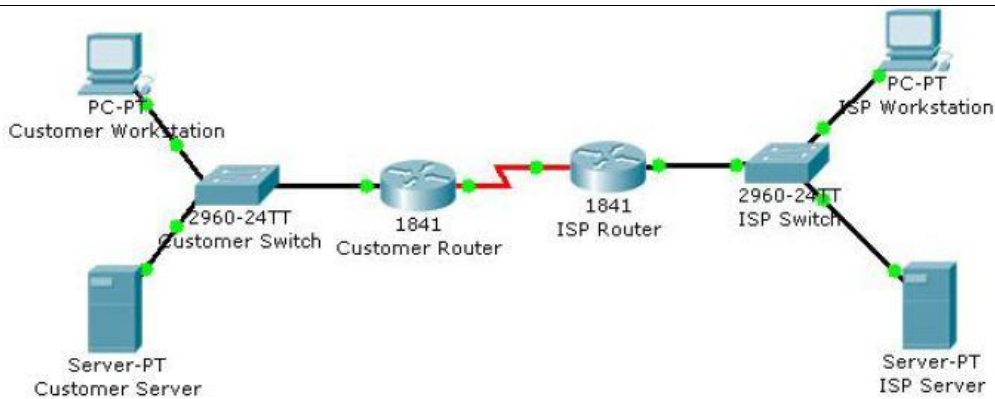
Step 1: Verify network connectivity

Required file: Placing ACLs

Step 1: Verify network connectivity

a. Verify that all of the PCs can communicate with each other and with the servers.

	b. Verify that the Internet Host can access the Web server (192.168.0.3), Sales server (192.168.10.2) and HR server (192.168.40.2) using the browser. Step 2: Examine the Access Control Lists that are configured on the routers a. Access the Distribution1 router. Use the following commands to view the ACL that has been configured on the Distribution1 router: □ show running-config □ show access-lists 1 b. Access the Gateway router. Use the following commands to view the ACL that has been configured on the Gateway router: □ show running-config □ show access-lists 100 Step 3: Determine the appropriate interface to apply the ACLs a. After examining the ACLs determine on which interface the ACLs should be applied b. The ACL must be applied to an interface or subinterface before it will affect the network traffic c. The extended ACL should be placed closest to the source and the standard ACL should be closest to the destination. d. Remember that only one ACL per port, per protocol, per direction is allowed. e. Apply the ACL to the appropriate interface or subinterface. Step 4: Examine the affects of the ACL a. Internet Host should be able to ping any device in the network, except HR1 or HR server. b. Internet Host should be able to access Web server (192.168.0.3) using the browser. c. Internet Host should not be able to access either the HR server (192.168.40.1) or Sales server (192.168.10.2) using the browser. d. HR2 should be able to access HR server (192.168.40.1) using ping or the browser. e. RandD2 should not be able to access HR server (192.168.40.1) using ping or the browser. Reflection 1. How can ACLs be used to control the flow of network traffic? 2. By default, what is always the last statement in an ACL?
6	To configure DHCP and HTTP server.



Objectives

- Configure the customer Cisco 1841 ISR as a DHCP server.

Background / Preparation

In this activity, you will continue to configure the Cisco 1841 ISR router for the customer network by configuring the DHCP service. The customer has several workstations that need to be automatically configured

with IP addresses on the local subnet and appropriate DHCP options to allow access to the Internet. The DHCP pool will use the 192.168.1.0/24 network but the first 49 addresses are excluded. The default

gateway and DNS server also need to be configured as 192.168.1.1 and 192.168.1.10.

For this activity, both the user and privileged EXEC passwords are cisco. Note: Packet Tracer does not currently support the domain name and lease period options. These options are not used in this activity.

Step 1: Configure the DHCP service.

a. From the customer workstation, use a console cable and terminal emulation software to connect to the console of the customer Cisco1841 ISR.

b. Log in to the console of the Cisco 1841 ISR and enter global configuration mode.

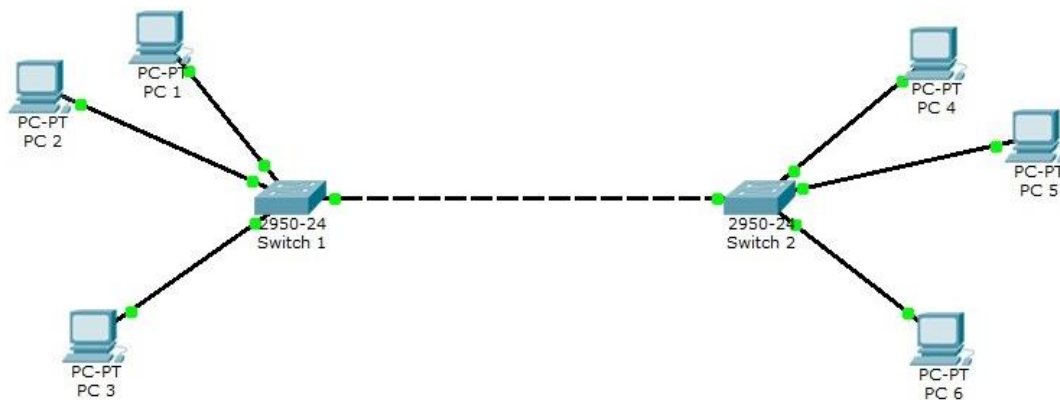
c. Before creating a DHCP pool, configure the addresses that are excluded. The range is from 192.168.1.1 to 192.168.1.49.

```
CustomerRouter(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.49
```

d. Create a DHCP pool called pool1.

```
CustomerRouter(config)#ip dhcp pool pool1
```

	e. Define the network address range for the DHCP pool. <pre>CustomerRouter(dhcp-config)#network 192.168.1.0 255.255.255.0</pre> f. Define the DNS server as 192.168.1.10. <pre>CustomerRouter(dhcp-config)#dns-server 192.168.1.10</pre> g. Define the default gateway as 192.168.1.1. <pre>CustomerRouter(dhcp-config)#default-router 192.168.1.1</pre> h. Add an exclusion range of 192.168.1.1 to 192.168.1.49 to the DHCP pool. <pre>CustomerRouter(dhcp-config)#exit CustomerRouter(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.49</pre> i. Exit the terminal. Step 2: Verify the DHCP configuration. a. From the customer workstation, open the Command Prompt window. b. Type <code>ipconfig /release</code> to release the current IP address. c. Type <code>ipconfig /renew</code> to request a new IP address on the local network. d. Verify that the IP address has been correctly assigned by pinging the LAN IP address of the Cisco 1841 ISR. e. Click the Check Results button at the bottom of this instruction window to check your work. Reflection a. What is the purpose of DHCP on the customer network? b. What IP address is assigned to the workstation after its IP address is renewed? c. What other DHCP options can be defined on the Cisco 1841 ISR router that are not configured in this activity?
7	Implement VLAN Configuration



Creating VLAN

In practice lab network Office1 Switch is configured as VTP Server. Office2 and Office3 switches are configured as VTP clients. We only need to create VLANs in VTP Server. VTP Server will propagate this information to all VTP clients automatically.

vlan **vlan number** command is used to create the VLAN.

Office 1 Switch

```
S1(config)#vlan 10
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#exit
S1(config)#
```

Assigning VLAN Membership

VLAN can be assigned statically or dynamically. CCNA exam only includes static method; therefore we will also use static method to assign VLAN membership. **switchport access** **vlan [vlan number]** command is used to assign VLAN to the interface. Following commands will assign VLANs to the interfaces.

Office 1 Switch

```
S1(config)#interface fastEthernet 0/1
S1(config-if)#switchport access vlan 10
S1(config-if)#interface fastEthernet 0/2
S1(config-if)#switchport access vlan 20
```

Office 2 Switch

```
S2(config)#interface fastEthernet 0/1
S2(config-if)#switchport access vlan 10
S2(config-if)#interface fastEthernet 0/2
S2(config-if)#switchport access vlan 20
```

Office 3 Switch

```
S3(config)#interface fastEthernet 0/1
S3(config-if)#switchport access vlan 10
S3(config-if)#interface fastEthernet 0/2
S3(config-if)#switchport access vlan 20
```

We have successfully assigned VLAN membership. It's time to test our configuration. To test this configuration, we will use **ping** command. **ping** command is used to test connectivity between two devices. As per our configuration, devices from same VLAN can communicate. Devices from different VLANs must not be able to communicate with each other without router.

Testing VLAN configuration

Access PC's command prompt to test VLAN configuration. Double click **PC-PT** and click **Command Prompt**



We have two VLAN configurations VLAN 10 and VLAN 20. Let's test VLAN 10 first. In VLAN 10 we have three PCs with IP addresses 10.0.0.2, 10.0.0.3 and 10.0.0.4. These PCs must be able to communicate with each other's. At this point PCs from VLAN 10 should not be allowed to access PCs from VLAN 20. VLAN 20 also has three PCs 20.0.0.2, 20.0.0.3 and 20.0.0.4.

```
PC>ipconfig
IP Address.....: 10.0.0.3
Subnet Mask.....: 255.0.0.0
Default Gateway...: 10.0.0.1
PC>ping 10.0.0.2
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=0ms TTL=128
PC>ping 10.0.0.4
Reply from 10.0.0.4: bytes=32 time=1ms TTL=128
Reply from 10.0.0.4: bytes=32 time=1ms TTL=128
PC>ping 20.0.0.4
Request timed out.
Request timed out.
PC>ping 20.0.0.3
Request timed out.
Request timed out.
PC>ping 20.0.0.2
Request timed out.
Request timed out.
```

We have successfully implemented VLAN 10 now test VLAN 20.

Same as VLAN 10, PCs from VLAN 20 must be able to communicate with other PCs of same VLAN while they should not be able to access VLAN 10.

```
PC>ipconfig
IP Address.....: 20.0.0.3
Subnet Mask.....: 255.0.0.0
Default Gateway...: 20.0.0.1
PC>ping 20.0.0.2
Reply from 20.0.0.2: bytes=32 time=34ms TTL=128
Reply from 20.0.0.2: bytes=32 time=0ms TTL=128
PC>ping 20.0.0.4
Reply from 20.0.0.4: bytes=32 time=12ms TTL=128
Reply from 20.0.0.4: bytes=32 time=0ms TTL=128
PC>ping 10.0.0.2
Request timed out.
Request timed out.
PC>ping 10.0.0.3
Request timed out.
Request timed out.
PC>ping 10.0.0.4
Request timed out.
Request timed out.
```

Congratulations we have successfully achieved one more mile stones of this article.

Configure Router on Stick

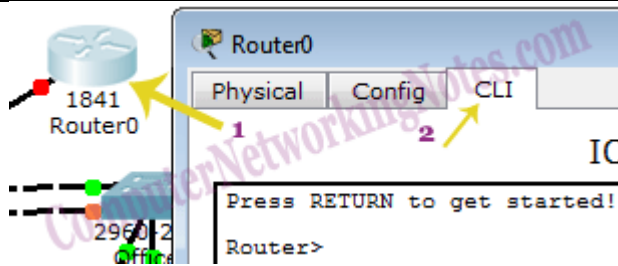
Typically routers are configured to receive data on one physical interface and forward that data from another physical interface based on its configuration. Each VLAN has a layer 3 address that should be configured as default gateway address on all its devices. In our scenario we reserved IP address 10.0.0.1 for VLAN 10 and 20.0.0.1 for VLAN 20.

With default configuration we need two physical interfaces on router to make this intra VLAN communication. Due to price of router, it's not a cost effective solution to use a physical interface of router for each VLAN. Usually a router has one or two Ethernet interface. For example if we have 50 VLANs, we would need nearly 25 routers in order to make intra VLANs communications. To deal with situation we use Router on Stick.

Router on Stick is router that supports trunk connection and has an ability to switch frames between the VLANs on this trunk connection. On this router, single physical interface is sufficient to make communication between our both VLANs.

Access command prompt of Router

To configure Router on Stick we have to access CLI prompt of Router. Click **Router** and Click **CLI** from menu items and Press **Enter key** to access the CLI



Run following commands in same sequence to configure Router on Stick

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastEthernet 0/0
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface fastEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 10.0.0.1 255.0.0.0
Router(config-subif)#exit
Router(config)#interface fastEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 20.0.0.1 255.0.0.0
Router(config-subif)#exit
```

In above configuration we broke up single physical interface [FastEthernet 0/0] into two logical interfaces, known as sub-interfaces. Router supports up to 1000 interfaces including both physical and logical.

By default interface link works as access link. We need to change it into trunk link. encapsulation commands specify the trunk type and associate VLAN with sub-interface.

In next step we assigned IP address to our sub-interface.

That's all configuration we need to switch VLANs. Now we can test different VLAN communications. To test intra VLANs communication open command prompt of PC and ping the PC of other VLAN.

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ipconfig
IP Address.....: 10.0.0.3
Subnet Mask.....: 255.0.0.0
Default Gateway...: 10.0.0.1

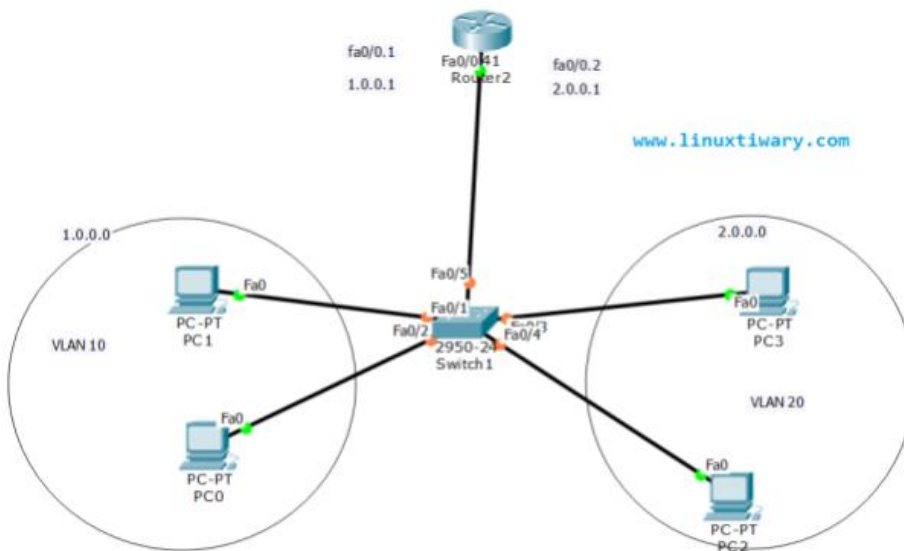
PC>ping 20.0.0.2
Reply from 20.0.0.2: bytes=32 time=0ms TTL=127
Reply from 20.0.0.2: bytes=32 time=0ms TTL=127
PC>
```

PC [10.0.0.3] from VLAN 10 can now access PC [20.0.0.2] from VLAN 20.

Spanning Tree Protocol (STP)

STP is a layer 2 protocol, used for removing loops. For backup purpose we typically create backup links for important resources. In our scenario, all offices have backup links that create loops in topology. STP automatically removes layer 2 loops. STP multicasts frame that contain information about switch interfaces. These frames are called BPDU (Bridge Protocol Data Units). Switch use BPDUs to learn network topology. If it found any loop, it will automatically remove that. To remove loop, STP disables port or ports that are causing it.

8 Inter VLAN Routing Configuration



```
SW1(config)#interface fa0/3
SW1(config-if)#switchport trunk encapsulation dot1q
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk allowed vlan 10,20
```

This is how we configure SW1. Make interface fa0/3 a trunk port and for security measures I made sure that only VLAN 10 and 20 are allowed.

```
R1(config)#interface fa0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.254 255.255.255.0
R1(config)#interface fa0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.254 255.255.255.0
```

Create two sub-interfaces on the router and tell it to which VLAN they belong. Don't forget to add an IP address for each VLAN.

```
R1#show ip route
```

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.10.0/24 is directly connected, FastEthernet0/0.10

C 192.168.20.0/24 is directly connected, FastEthernet0/0.20

The router will be able to route because these two networks are directly connected.

C:\Documents and Settings\H1>**ipconfig**

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address. : 192.168.10.1
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.10.254

C:\Documents and Settings\H2>**ipconfig**

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address. : 192.168.20.1
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.20.254

Don't forget to set your IP address and gateway on the computers.

Let's try a ping:

C:\Documents and Settings\H1>**ping 192.168.20.1**

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=128
Reply from 192.168.20.1: bytes=32 time<1ms TTL=128
Reply from 192.168.20.1: bytes=32 time<1ms TTL=128
Reply from 192.168.20.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:

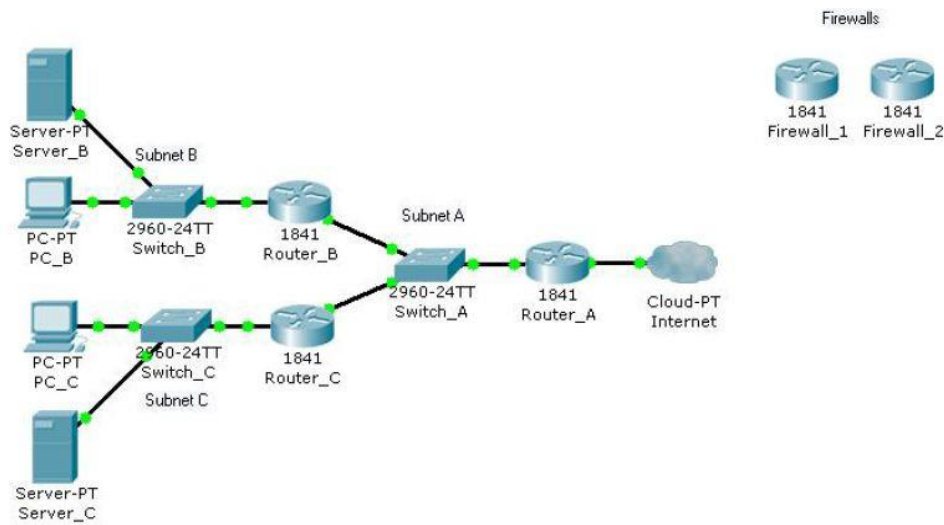
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

That's how you do it. So why would you want to use a solution like this? It's cheap! You don't need a multilayer switch for your routing. Any layer 2 switch will do.

The Cisco Catalyst 2960 is a layer 2 switch; the cheapest multilayer switch is the Cisco Catalyst 3560. Compare the price on those two and you'll see what I'm talking about.

Some of the disadvantages of this solution is that your router is a single point of failure and that traffic flows up and down on the same link which might cause congestion.

9 Planning Network-based Firewalls



Objectives

- Place firewalls in appropriate locations to satisfy security requirements.

Background / Preparation

You are a technician who provides network support for a medium-sized business. The business has grown and

includes a research and development department working on a new, very confidential project. The livelihood of

the project depends on protecting the data used by the research and development team.

Your job is to install firewalls to help protect the network, based on specific requirements. The

Packet Tracer

topology that you will use includes two preconfigured firewalls. In the two scenarios presented, you will replace

the existing routers with the firewalls. The firewalls need to be configured with the appropriate IP address

configurations, and the firewalls should be tested to ensure that they are installed and configured correctly.

Scenario 1: Protecting the Network from Hackers

Because the company is concerned about security, you recommend a firewall to protect the network from

hackers on the Internet. It is very important that access to the network from the Internet is restricted.

Firewall_1 has been preconfigured with the appropriate rules to provide the security required. You will install it on the network and confirm that it is functioning as expected.

Step 1: Replace Router_A with Firewall_1.

a. Remove Router_A and replace it with Firewall_1. b. Connect the Fast Ethernet 0/0 interface on Firewall_1 to the Fast Ethernet 0/1 interface on Switch_A. Connect the Fast Ethernet 0/1 interface on Firewall_1 to the Ethernet 6 interface of the ISP cloud. (Use straight-through cables for both connections.) c. Confirm that the host name of Firewall_1 is Firewall_1. d. On Firewall_1, configure the WAN IP address and subnet mask for the FastEthernet 0/1 interface as 209.165.200.225 and 255.255.255.224. e. Configure the LAN IP address and subnet mask for the Fast Ethernet 0/0 interface on Firewall_1

as 192.168.1.1 and 255.255.255.0. Step 2: Verify the Firewall_1 configuration.

a. Use the show run command to verify your configuration. This is a partial example of the output.

```
Firewall_1#show run
```

```
Building configuration...
```

```
hostname Firewall_1
```

```
!
```

```
interface FastEthernet0/0
```

```
ip address 192.168.1.1 255.255.255.0
```

```
ip nat inside
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface FastEthernet0/1
```

```
ip address 209.165.200.225 255.255.255.224
```

```
ip access-group 100 in
```

```
ip nat outside
```

```
duplex auto
```

```
speed auto
```

```
!
```



```
interface Vlan1
no ip address
shutdown
!
ip nat inside source list 1 interface FastEthernet0/0 overload
ip classless
ip route 192.168.2.0 255.255.255.0 192.168.1.2
ip route 192.168.3.0 255.255.255.0 192.168.1.3
!
access-list 1 permit 192.168.0.0 0.0.255.255
access-list 100 deny ip any host 209.165.200.225
<output omitted>
!
end
```

b. From PC_B, ping 209.165.200.225 to verify that the internal computer can access the Internet.

PC>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time=107ms TTL=120

Reply from 209.165.200.225: bytes=32 time=98ms TTL=120

Reply from 209.165.200.225: bytes=32 time=104ms TTL=120

Reply from 209.165.200.225: bytes=32 time=95ms TTL=120

Ping statistics for 209.165.200.225:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 95ms, Maximum = 107ms, Average = 101ms

c. From privileged EXEC mode on Firewall_1, save the running configuration to the startup configuration

using the copy run start command. Scenario 2: Securing the Research and Development Network

Now that the entire network is secured from traffic originating from the Internet, secure the research and

development network, Subnet C, from potential breaches from inside the network. The research and

development team needs access to both the server on Subnet B and the Internet to conduct research.

Computers

on Subnet B should be denied access to the research and development subnet.

Firewall_2 has been preconfigured with the appropriate rules to provide the security required. You will install it

on the network and confirm that it is functioning as expected.

Step 1: Replace Router_C with Firewall_2.

a. Remove Router_C and replace it with Firewall_2. b. Connect the Fast Ethernet 0/1 interface on Firewall_2 to the Fast Ethernet 0/3 interface on

Switch_A. Connect the Fast Ethernet 0/0 interface on Firewall_2 to the Fast Ethernet 0/1 interface on Switch_C. (Use straight-through cables for both connections.)

c. Confirm that the host name of Firewall_2 is Firewall_2. d. On Firewall_2, configure the WAN IP address and subnet mask for the Fast Ethernet 0/1 interface

as 192.168.1.3 and 255.255.255.0. e. Configure the LAN IP address and subnet mask for the Fast Ethernet 0/0 interface of Firewall_2

as 192.168.3.1 and 255.255.255.0. Step 2: Verify the Firewall_2 configuration.

a. Use the show run command to verify the configuration. This is a partial example of the output.

```
Firewall_2#show run
```

```
Building configuration...
```

```
... !
```

```
interface FastEthernet0/0
```

```
ip address 192.168.3.1 255.255.255.0
```

```
ip nat inside
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface FastEthernet0/1
```

```
ip address 192.168.1.3 255.255.255.0
```

```
ip access-group 100 in
```

```
ip nat outside
```

```
duplex auto
```

speed auto

!

access-list 1 permit 192.168.3.0 0.0.0.255

access-list 100 permit ip host 192.168.2.10 any

access-list 100 permit ip host 192.168.1.1 any

<output omitted>

!

end

b. From the command prompt on PC_B, use the ping command to verify that the computers on Subnet B

cannot access the computers on Subnet C.

PC>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.3.10:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

c. From the command prompt on PC_C, use the ping command to verify that the computers on Subnet C can

access the server on Subnet B.

PC>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.

Reply from 192.168.2.10: bytes=32 time=164ms TTL=120

Reply from 192.168.2.10: bytes=32 time=184ms TTL=120

Reply from 192.168.2.10: bytes=32 time=142ms TTL=120

Ping statistics for 192.168.2.10:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

Minimum = 142ms, Maximum = 184ms, Average = 163ms

d. From the command prompt on PC_C, use the ping command to verify that the computers on Subnet C can access the Internet.

```
PC>ping 209.165.200.225
```

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time=97ms TTL=120

Reply from 209.165.200.225: bytes=32 time=118ms TTL=120

Reply from 209.165.200.225: bytes=32 time=100ms TTL=120

Reply from 209.165.200.225: bytes=32 time=110ms TTL=120

Ping statistics for 209.165.200.225:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 97ms, Maximum = 118ms, Average = 106ms

e. From privileged EXEC mode on Firewall_2, save the running configuration to the startup configuration

using the copy run start command. f. Click the Check Results button at the bottom of this instruction window to check your work.

Reflection

a. Why would you install a firewall on the internal network?

b. How does a router that is configured to use NAT help protect computer systems on the inside of the NAT router?

c. Examine the location of Firewall_1 and Firewall_2 in the completed network topology. Which networks are considered trusted and untrusted for Firewall_1? Which networks are considered trusted and untrusted for Firewall_2?